MECHATRONIC SYSTEM INTEGRATION
(MCTA 3203)

SEMESTER 1, 2026/2027

EXPERIMENT 3

DESIGNING A MOTION-ACTIVATED RFID SYSTEM

MCTA 3203

SECTION 1

GROUP 8

NO.	MATRIC NO.	NAME
1	2318483	SYED MOHAMAD SYAFIEE BIN SYED TAJUDIN
2	2413869	SHARIF NUR AZWAR BIN SHARIF ABU BAKAR
3	2413733	ZAHIN ZAYDAN BIN ZAIRY

SUBMISSION DATE : 11 OCTOBER 2026

Abstract

This experiment focuses on understanding digital logic systems and the practical interfacing of a 7-segment display using a Cytron CT-UNO microcontroller. The main objective is to learn how to control a 7-segment display using digital output signals and push buttons, while applying basic digital logic concepts in a hardware system. A common cathode 7-segment display was connected to the CT-UNO through digital output pins, allowing the microcontroller to control individual segments to display numeric values from 0 to 9. The system was programmed to increase the displayed number by one whenever the increment push button was pressed. When the number reached 9, the next button press returned the display to 0, creating a continuous counting sequence. A separate reset push button was also implemented to immediately reset the displayed number to 0. In addition, debounce handling was incorporated into the program to prevent multiple counts from being registered due to unwanted electrical signals generated when the buttons were pressed. Throughout this experiment, the relationship between software programming and hardware operation was explored. The experiment successfully demonstrated the application of digital outputs, input signals, and basic control logic in a microcontroller-based system. Overall, this experiment improved the understanding of digital interfacing and provided practical experience in designing and implementing a simple electronic counting system.

Table of Content

Abstract

1. Introduction

2. Material and Equipment

3. Experiment Setup

4. Methodology

5. Data Collection

6. Data Analysis

7.Result

8. Conclusion

9. Recommendations

10. References

11. Appendices

12. Acknowledgement

13. Student Declaration

Week 2: Digital Logic System: Basic Logic Gates, Electronic Circuit Interfacing, Basic ALU, 7 segment display, ICs based interfacing application

1. Introduction

This week's laboratory session focused on Digital Logic Systems, which form the fundamental basis of digital electronic devices, including computers. In digital systems, signals are represented by two binary states: 1 (HIGH or ON) and 0 (LOW or OFF). The experiment introduced the basic concepts of logic gates, interfacing techniques, and digital display devices. In particular, an Arduino Uno board was used to control a 7-segment display, allowing numerical digits to be displayed through digital signals. The pin configuration of the 7-segment display is illustrated in **Figure 1.0** below.

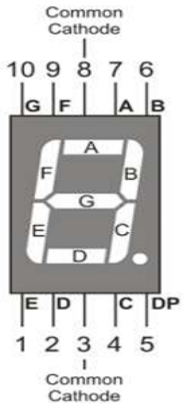


Figure 1.0 Shows the 7-segment display pinout

The main objectives of this experiment were as follows:

- To understand the operating principle of a 7-segment display and learn how to control its individual segments.
- To develop practical skills in connecting electronic components using a breadboard.
- To understand how digital outputs are programmed to display specific numerical digits.
- To implement and control the number increment sequence and reset function using push

buttons.

- To ensure the 7-segment display shows numbers from 0 to 9 and the push buttons function correctly for incrementing and resetting.

2. Materials and Equipments

● Cytron CT-UNO	1
● Common cathode 7-segment display	1
● 220 Ω resistors	8
● 10 kΩ resistors	2
● Pushbuttons	2
● Breadboard	1
● Micro USB-Cable	1
● Jumper wires	

3. Experimental Setup

A 7-segment display consists of seven LEDs labeled A to G. By activating different combinations of these segments, the display can represent numbers from 0 to 9.The common cathode was connected to 5 V instead of GND, reversing the logic operation. Therefore, the segments turn ON when the Cytron UNO output is LOW and turn OFF when it is HIGH. Segments A to G were connected to digital pins D2 to D9 through 220 Ω resistors to limit current and protect the components.

Two pushbuttons were also connected:

- **Button 1 (Increment):** Connected to pin D10
- **Button 2 (Reset):** Connected to pin D11

Both push buttons were connected to GND through 10 kΩ resistors to maintain stable logic levels. All connections were connected based on circuit **Figure 2.0**.

Circuit Connection:

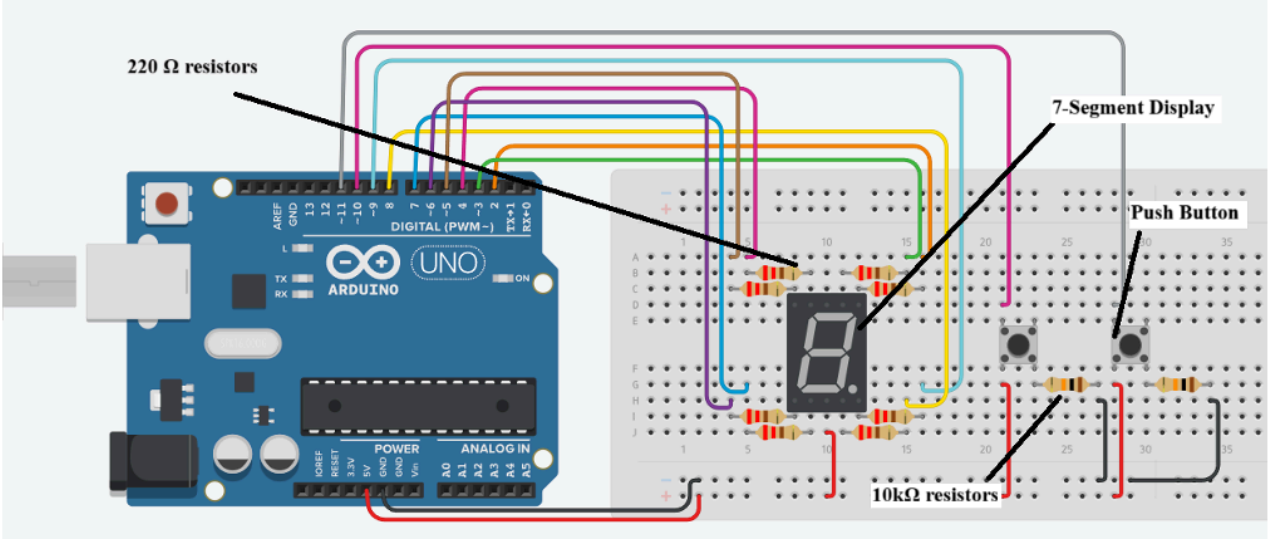


Figure 2.0 Show the schematic diagram based on ThinkerCad

- A HIGH signal from the Cytron UNO turns the segment OFF.
- A LOW signal activates the segment.
- Different combinations of activated segments form the numbers displayed.

4. Methodology

4.1 Circuit Assembly

All components were connected on a breadboard according to Figure 2.0. The seven-segment display pins (A–G) were connected to the Cytron UNO digital pins D2–D9 through 220 Ω resistors. The common cathode pin was connected to 5 V, reversing the logic so that LOW activates the segments while HIGH deactivates them.

4.2 Button Connection

The increment and reset buttons were connected to digital pins D10 and D11, respectively. Each button was connected to GND through a 10 kΩ resistor and to 5 V on the opposite side.

4.3 Code Upload

The Arduino IDE was used to program and upload the code to the Cytron UNO, controlling the segments to display numbers from 0 to 9.

4.3.1 Example Coding (Continuous Automatic)

The code was programmed to display numbers from 0 to 9 in sequence continuously, with a 1-second delay between each number.

```
// Define the pins for each segment (D0 to D6)
const int segmentA = 0; // D0
const int segmentB = 1; // D1
const int segmentC = 2; // D2
const int segmentD = 3; // D3
const int segmentE = 4; // D4
const int segmentF = 5; // D5
const int segmentG = 6; // D6

void setup() {
  // Initialize the digital pins as OUTPUTs
  pinMode(segmentA, OUTPUT);
  pinMode(segmentB, OUTPUT);
  pinMode(segmentC, OUTPUT);
  pinMode(segmentD, OUTPUT);
  pinMode(segmentE, OUTPUT);
  pinMode(segmentF, OUTPUT);
  pinMode(segmentG, OUTPUT);
}

// 0 = A,B,C,D,E,F
// 1 = B,C
// 2 = A,B,G,E,D
// 3 = A,B,C,D,G
// 4 = A,F,G,C,D

void loop() {
  // Turn on each segment one by one
  digitalWrite(segmentA, HIGH);
  digitalWrite(segmentB, HIGH);
  digitalWrite(segmentC, HIGH);
  digitalWrite(segmentD, HIGH);
  digitalWrite(segmentE, HIGH);
  digitalWrite(segmentF, HIGH);

delay(500);
digitalWrite(segmentA, LOW);
digitalWrite(segmentB, LOW);
digitalWrite(segmentC, LOW);
digitalWrite(segmentD, LOW);
digitalWrite(segmentE, LOW);
digitalWrite(segmentF, LOW);

delay(500);
digitalWrite(segmentB, HIGH);
digitalWrite(segmentC, HIGH);
delay(500);
digitalWrite(segmentB, LOW);
digitalWrite(segmentC, LOW);

delay(500);
digitalWrite(segmentA, HIGH);
digitalWrite(segmentB, HIGH);
digitalWrite(segmentG, HIGH);
digitalWrite(segmentE, HIGH);
digitalWrite(segmentD, HIGH);
delay(500);
digitalWrite(segmentA, LOW);
digitalWrite(segmentB, LOW);
digitalWrite(segmentG, LOW);
digitalWrite(segmentE, LOW);
digitalWrite(segmentD, LOW);

delay(500);
digitalWrite(segmentA, HIGH);
digitalWrite(segmentB, HIGH);
digitalWrite(segmentC, HIGH);
digitalWrite(segmentD, HIGH);
digitalWrite(segmentG, HIGH);
delay(500);
digitalWrite(segmentA, LOW);
digitalWrite(segmentB, LOW);
digitalWrite(segmentC, LOW);
digitalWrite(segmentD, LOW);
digitalWrite(segmentG, LOW);

  delay(500);
digitalWrite(segmentA, HIGH);
digitalWrite(segmentF, HIGH);
digitalWrite(segmentG, HIGH);
digitalWrite(segmentC, HIGH);
digitalWrite(segmentD, HIGH);
delay(500);
digitalWrite(segmentA, LOW);
digitalWrite(segmentF, LOW);
digitalWrite(segmentG, LOW);
digitalWrite(segmentC, LOW);
digitalWrite(segmentD, LOW);

delay(500); // Delay for 1/2 second before repeating
}
```

4.3.1 Task Coding (Increment)

The code was programmed to display a number on the 7-segment display, starting from 0 when the system is powered on. Each time the increase push button is pressed, the displayed number increases by 1. When the count reaches 9, the next press returns the display to 0. Meanwhile, pressing the reset push button immediately resets the display to 0, regardless of the current number. This allows the user to count repeatedly from 0 to 9 using the increase button and reset the count whenever necessary.

```
const int segmentA = 3;
const int segmentB = 2;
const int segmentC = 8;
const int segmentD = 7;
const int segmentE = 6;
const int segmentF = 4;
const int segmentG = 5;
const int button_1 = 10;
const int button_2 = 11;

int count = 0;
int num_array[10][7] = {
  {1, 1, 1, 1, 1, 1, 0},//0
  {0, 1, 1, 0, 0, 0, 0},//1
  {1, 1, 0, 1, 1, 0, 1},//2
  {1, 1, 1, 1, 0, 0, 1},//3
  {0, 1, 1, 0, 0, 1, 1},//4
  {1, 0, 1, 1, 0, 1, 1},//5
  {1, 0, 1, 1, 1, 1, 1},//6
  {1, 1, 1, 0, 0, 0, 0},//7
  {1, 1, 1, 1, 1, 1, 1},//8
  {1, 1, 1, 1, 0, 1, 1} //9
};

void setup() {
  pinMode(segmentA, OUTPUT);
  pinMode(segmentB, OUTPUT);
  pinMode(segmentC, OUTPUT);
  pinMode(segmentD, OUTPUT);
  pinMode(segmentE, OUTPUT);
```

```
  pinMode(segmentF, OUTPUT);
  pinMode(segmentG, OUTPUT);

  pinMode(button_1, INPUT);
  pinMode(button_2, INPUT);

  display(num_array[count]);
}

void loop() {
  // Check Increment Button (Button 1 on Pin 10)
  if (digitalRead(button_1) == LOW) {
    if (count < 9) {
      count++;
    } else {
      count = 0; // Loop back to 0 after 9
    }
    display(num_array[count]);
    delay(200); // Debounce delay
  }

  // Check Reset Button (Button 2 on Pin 11)
  if (digitalRead(button_2) == LOW) {
    count = 0; // Reset count to 0
    display(num_array[count]);
    delay(200); // Debounce delay
  }
}

void display(int bin[]) {
  digitalWrite(segmentA, bin[0]);
  digitalWrite(segmentB, bin[1]);
  digitalWrite(segmentC, bin[2]);
  digitalWrite(segmentD, bin[3]);
  digitalWrite(segmentE, bin[4]);
  digitalWrite(segmentF, bin[5]);
  digitalWrite(segmentG, bin[6]);
}
```

5. Data Collection

Result is collected through observation by our own eyes

1. The LED lights up when the code is put as HIGH.
2. The LED lights off for 0.5 seconds when delay 500 is put.
3. Push button is triggered when the code is put as HIGH.

6. Data Analysis

The data confirmed that:

1. The 7-Segment display is working well.
2. The wire connection is working well.
3. The pushbutton is working well
4. The code is correctly written and working fine.

7. Results and Discussions

The Results indicated a successful required data collection, the arduino uno code was written well to display the numbers correctly on the 7-segment display. The first push-button is used to increment the number from 0 to 9 flawlessly. The second push-button is used to reset the counter to zero and also works flawlessly. To avoid burnt display, resistors are put in the connection to reduce current that would burn the display.

Questions: How can you interface an I2C LCD with Arduino? Explain the coding principle behind it compared to a 7-segment display and a matrix LED.

Compared to I2C LCD, 7-segment display is rather easier to code because for 7-segment display, code required is only to put high or low for each segment to indicate whether each segment is on or off. While for I2C LCD, the code is similar to other languages; by typing “print” followed by the texts to display on the I2C LCD. It also needs to be turned on by initializing the variables to turn on the LCD and the backlight. Subsequently, the position of text will be coded followed by the texts of the LCD.

CONCLUSION

The experiment successfully achieved its objective of understanding digital logic system and practical hardware interfacing using a Cytron CT-UNO microcontroller and a 7 segment display.The arduino IDE code was successfully compiled and uploaded to control individual segments to display numeric values from 0 to 9 sequentially. The push button increment worked perfectly to increase the number count sequentially and loop back to 0 after reaching 9, while the reset push button successfully reset the display back to 0 . The use of 220Ω and 10 kΩ resistors effectively limited current to prevent component damage and maintained stable logic signal levels.

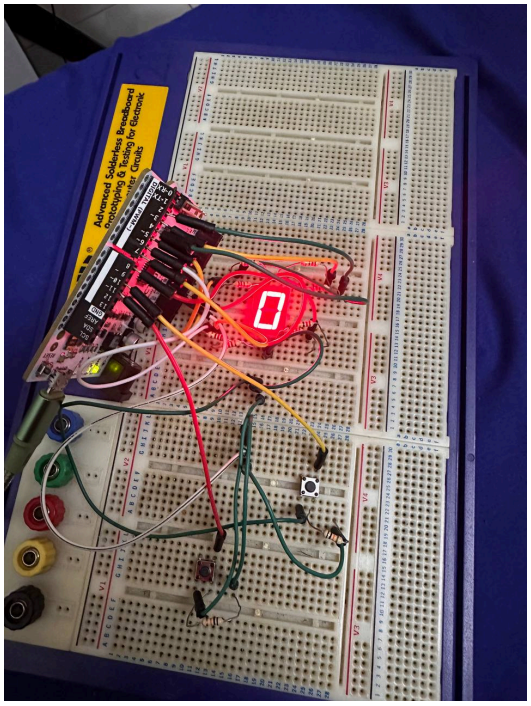
RECOMMENDATIONS

For future improvements, it is recommended to replace the single 7 segment display with an 12C lCD or LED matrix module to allow multi character text and more comprehensive information display. Secondly,replace the jumper wires with the longer one to make the circuit more neater and well organized.Lastly, modify the code and adding the third push button to enable decrement function.

REFERENCES

How to Wire and Program a Button | Arduino Documentation. (2024, October 2). Docs.Arduino.Cc. https://docs.arduino.cc/built-in-examples/digital/Button/

APPENDICES



Link for video simulation

https://drive.google.com/file/d/1DyyPI-609eXAFimMzmY7zJxQMyQiHqw8/view?usp=drive_link

ACKNOWLEDGMENTS

Alhamdulillah, all praise and gratitude be to Allah SWT for His blessings, guidance, and mercy that enabled us to successfully complete this experiment.Throughout the development process, we faced several challenges, but with His will and
guidance, we managed to overcome them and complete the project successfully.
We would also like to express our heartfelt appreciation to our lecturer for their continuous guidance, support, and valuable feedback throughout this project. Their advice has greatly helped us in understanding both the theoretical and practical of digital logic systems and practical hardware interfacing using a Cytron CT-UNO microcontroller and a 7 segment display.

Student's Declaration

We hereby declare that the work presented in this technical report is entirely our own, except where explicitly acknowledged. We certify that in developing and completing this mechatronics project, we have strictly adhered to the standards of academic integrity and have not engaged in any form of plagiarism or unethical conduct. All sources of information, reference materials, and technical support utilized in this work have been appropriately cited and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.We therefore, agreed unanimously that this report shall be submitted for marking and this final
The printed report has been verified by us.

Signature: sharif	Read [/]
Name: Sharif Nur Azwar Bin Sharif Abu Bakar	Understand [/]
Matric Number: 2413869	Agree [/]
Signature: syed	Read [/]
Name: Syed Mohammad Syafiee Bin Syed Tajudin	Understand [/]
Matric Number:2318483	Agree [/]
Signature: zahin	Read [/]
Name: Zahin Zaydan bin Zairy	Understand [/]

Matric Number: 2413733 Agree [/]